

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

MODERN COMPUTER VISION [NPT-4078]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given the following sets of lines, compute the point of intersection for each pair of lines using homogeneous representation (4)
 - A) a) $x - 3y = 0$ and $4x + 5y = 0$,
b) $x = 6$ and $x = 7$.
 - B) Suppose we have a television view of a soccer field with players moving around. Assuming that each player occupies 10-30 pixels rectangular box. Explain a procedure to track the rectangular box in the video assuming a pure translation motion model. (3)
 - C) Describe the correlation-based method for finding the pixel-wise image correspondence. State one limitation of the correlation-based approaches for binocular fusion. (3)
- 2) Compare Uniform and Gaussian smoothing. Discuss the effect of varying the standard deviation of the Gaussian kernel in Gaussian smoothing for noise removal. (4)
 - A)
 - B) Assume the camera parameters (camera projection matrices) and the image points x, x' corresponding to the scene point X are known. Explain two triangulation methods for computing scene point X . (3)
 - C) Define Quaternions. Explain how Quaternions can be used to compute the rigid transformation for registering range images. (3)
- 3) (5)
 - A)

Given two 8-bit grayscale images shown in Figure below, calculate the joint probability $p(x, y)$ of the two images. With the help of these two images as an example, describe the process of image registration using mutual information.

0	0	0	0	0
	20	20	20	0
0				
0	20	20	20	0
0	0	0	0	0
0	0	0	0	0

Image 1

0	0	0	0	0
0	0	0	0	0
0	20	20	20	0
0	20	20	20	0
0	0	0	0	0

Image 2

- B) Show that for a video $f(x, y, t)$, the brightness constancy assumption can be expressed as $f_x u + f_y v + f_t = 0$, where f_x, f_y, f_t are the partial derivatives of $f(x, y, t)$ wrt x, y , and t respectively, and u, v is the optical flow vector components. Explain the algorithm to compute Horn and Schunck's optical flow and derive the update equation. (5)
- 4) Given two uncalibrated cameras, explain the steps to compute a projective reconstruction of the scene from image correspondence. (4)
- A) (3)
- B) (3)

Given a rigid object, where the target set ($T = \{y_j\}$, $j=1, \dots, N$) is rotated, translated and scaled version of the source set ($S = \{x_i\}$, $i=1, \dots, M$) and there might be some noise. Explain a least square based procedure for estimating the rotation, translation and scale parameter(s).

C)

Describe roof edges in range images. Discuss the additional information included in the range images that makes it more suitable for 3D reconstruction task. (3)

5) Explain the algebraic derivation of the fundamental matrix. (5)

A)

B) Explain affine epipolar geometry and affine epipolar constraint. Assume that the affine fundamental matrix has been computed, explain the procedure to compute the camera matrix from the affine fundamental matrix using the affine epipolar constraint. (5)

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